

Ahmed Messoudi

AI & Data Engineer | Data Engineering Student — Faculty of Sciences of Sfax
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PROFESSIONAL PROFILE

Second-year **Data Engineering** student at the Faculty of Sciences of Sfax, designing end-to-end solutions spanning machine learning, deep learning, microservices, DevOps, and data warehousing — from conception through production deployment.

EDUCATION

Engineering Degree in Data Engineering — *Faculty of Sciences of Sfax* 2024 – Present
Algorithms, Advanced Databases, Machine Learning, Distributed Systems, Big Data, Signal Processing, Statistics, Networking.

Integrated Preparatory Classes — *Faculty of Sciences of Sfax* 2022 – 2024
Mathematics, Physics, Fundamentals of Computer Science, Linear Algebra, Analysis, Algorithmics.

Baccalaureate — Science Track — *High School, Monastir* 2022

TECHNICAL SKILLS

Languages	Python, Java, JavaScript / TypeScript, SQL, Bash
Machine Learning	Scikit-learn, XGBoost, Random Forest, SVM, Feature Engineering, MLflow
Deep Learning	TensorFlow, PyTorch, LSTM, CNN, Transformers (BERT), ECAPA-TDNN, Transfer Learning
NLP & AI	NLTK, SpaCy, HuggingFace, Sentiment Analysis, Text Classification, OCR (Tesseract, Easy-OCR)
Data Engineering	Apache Spark, Kafka, Airflow, Pandas, ETL, Data Warehousing (SSIS, SSAS), Pentaho
Databases	PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch
DevOps & Cloud	Docker, Kubernetes, Jenkins, Ansible, Vagrant, CI/CD, AWS
Microservices	Spring Boot, Flask, Node.js, FastAPI, REST API
Frontend	React, Angular, TypeScript, Tailwind CSS
Visualization	Kibana, Power BI, Matplotlib, Plotly
Tools	Git, GitHub, GitLab, VS Code, Jupyter

ACADEMIC & PERSONAL PROJECTS

AVC Micro-Services — Multimodal Stroke Detection AI [github]

Multimodal stroke detection system combining a CNN (EfficientNetB0 + Grad-CAM) for CT scan classification, an NLP engine (SpaCy + XGBoost) for risk factor extraction, and a vision-language model (Qwen2-VL) for structured biological report analysis. All three AI components are integrated into a FastAPI backend deployed as Kubernetes-orchestrated microservices with horizontal autoscaling.

Stack: Python, PyTorch, EfficientNetB0, XGBoost, SpaCy, FastAPI, Docker, Kubernetes.

DevOps & Kubernetes Cluster [github]

Fully infrastructure-as-code DevOps environment (Vagrant + Ansible) deploying a multi-node Kubernetes cluster with Calico CNI, persistent NFS storage, and a private Docker registry (Nexus). A Jenkins CI/CD pipeline running as a Kubernetes pod with a Docker-in-Docker sidecar automates build, test, publish to Nexus, and deployment with zero manual intervention.

Stack: Kubernetes, Docker, Jenkins, Ansible, Vagrant, Nexus, Calico CNI, Node.js.

Deep Speaker Verification System [github]

Speaker identity verification system benchmarking x-vector (TDNN) vs. ECAPA-TDNN architectures, trained with AAM-Softmax loss and rigorously evaluated on VoxCeleb under noisy and clean conditions. Both models are served through a containerized FastAPI exposing enrollment, verification, and identification endpoints.

Stack: Python, PyTorch, VoxCeleb, FastAPI, Docker, AAM-Softmax.

Tunisair Data Warehouse — Multi-source DWH [github]

Decision-support data warehouse for Tunisair integrating heterogeneous sources (JSON, XLSX, CSV, Access) through a Python + Talend ETL pipeline into a dimensional model centered on flights, airports, and dates. OLAP cubes (SSAS/Pentaho) feed Power BI dashboards answering key analytical questions on delays, passenger trends, and financial performance.

Stack: SQL Server, SSIS, SSAS, Pentaho, Talend, Power BI, Python.

MediFlow — Hospital Management System [github]

Full-stack hospital resource management application covering beds, wards, equipment, and patients, with four-level RBAC and an intelligent allocation engine that automatically prioritizes ICU beds for critical cases. Real-time capacity alerts and an occupancy KPI dashboard provide proactive resource management across all departments.

Stack: Spring Boot 3, Angular 17, PostgreSQL, Tailwind CSS, Chart.js, Docker, JWT.

Flow — Project & Client Management Platform [github]

Enterprise full-stack platform with multi-tenant architecture, five-level RBAC, a real-time client portal, versioned document management with drag-and-drop upload, and an integrated calendar with meeting requests. The Spring Boot REST API is secured by JWT; the React 18 frontend leverages React Query and Zod for server state management and validation.

Stack: Spring Boot 3, React 18, TypeScript, MySQL, JWT, Docker, Tailwind CSS.

AI Sentiment Analyzer — NoSQL Analysis of AI Tools [github]

Distributed sentiment analysis pipeline (TextBlob + VADER, weighted score) applied to reviews of popular AI tools including ChatGPT, Claude, Midjourney, and GitHub Copilot, stored in MongoDB with a flexible schema. Data is indexed in Elasticsearch with real-time aggregations and cross-tool sentiment comparisons visualized through Kibana dashboards.

Stack: Python, TextBlob, VADER, MongoDB, Elasticsearch, Kibana, Docker Compose.

League of Legends — Competitive Data Analysis [github]

Statistical analysis of ranked matches applying PCA, chi-squared tests, Mann-Whitney, and Lorenz curves to uncover win-rate patterns and identify undervalued “Hidden OP” champions overlooked by the meta. The study confirms that first inhibitor destruction is the most robust predictor of victory (supervised academic project).

Stack: Python, Pandas, SciPy, Scikit-learn, Matplotlib, Seaborn, Jupyter.

Global Startup Advisor — AI Market Intelligence [github]

Entrepreneurial decision-support platform combining a Data Mining pipeline on CIA World Factbook indicators, K-Means clustering (PCA-visualized), and supervised models (Random Forest, XGBoost) to assess startup viability by country and sector. A natural language interface via the Groq API lets users describe their project and receive a structured recommendations report covering Fintech, Edtech, and E-commerce sectors.

Stack: Python, Scikit-learn, XGBoost, K-Means, PCA, Plotly, Streamlit, Groq API.

Document Management System with OCR

Intelligent archiving application automating PDF and image ingestion, text extraction (Tesseract, EasyOCR), content structuring, and full-text Elasticsearch indexing for fast keyword and metadata search. A React administration dashboard manages documents, visualizes extracted metadata, and supports complex queries — the entire stack deployable with a single `docker-compose up`.

Stack: Python, Tesseract, EasyOCR, Elasticsearch, PostgreSQL, Flask, React, Docker.

LANGUAGES & SOFT SKILLS

Arabic	Native language	• Complex problem-solving & analytical thinking • Teamwork, Agile methodologies (Scrum/Kanban) • Self-driven, initiative & multi-project management • Technical communication & documentation writing • Technology watch & continuous learning
French	Fluent (academic writing, technical reports)	
English	Professional (documentation, GitHub)	